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Introduction

1.1 Motivation

Rapid progress in computational power, transformer architectures, and large-scale foundation models has transformed what robots can perceive, reason about, and do (Firoozi et al., 2025). Large language models (LLMs), and their multimodal counterparts (MLLMs) as well as vision-language-action models (VLAs), now enable robots to interpret open-ended instructions, plan multi-step actions (e.g., Ahn et al., 2023; Driess et al., 2023), and generate motor commands from visual and linguistic input (Black et al., 2025). As a result, robots have moved from isolated industrial cells into spaces shared with humans: surgical systems assist in operating rooms (Morrell et al., 2021), automated vehicles navigate urban traffic (Baek et al., 2024), socially assistive companions support elder care (e.g., Abdi et al., 2018; Yen et al., 2024), and LLM-driven planners have begun to direct Mars rover traversals (Anthropic, 2026; NASA JPL, 2026). This trend is driven by the ability of robots to increase labor productivity (Graetz & Michaels, 2018), reduce work-related physical injuries (Gihleb et al., 2022), and mitigate labor shortages caused by aging populations (Acemoglu & Restrepo, 2022). Yet, technical capability alone does not guarantee effective collaboration with humans who must supervise, instruct, or work together with these systems.

Fitts (1951) documented that humans and machines excel at fundamentally different aspects of information processing, and that observation remains important: humans bring flexible, context-sensitive perception and judgment, while artificial systems offer speed, consistency, and tireless computation. Modern LLMs have partially narrowed this divide by demonstrating general in-context information-processing, yet their errors can sound fluent and convincing, leading users to over-rely on incorrect responses (Kim et al., 2025). When a robot takes over more decision-making, the human role changes from active operator to monitor, supervisor or collaborator (Sheridan & Verplank, 1978), often degrading situation awareness and the ability to intervene when failures occur (Endsley & Kiris, 1995).

At the same time, new communication modalities open up: humans may direct robots through speech, gesture, or high-level prompting (Goodrich & Schultz, 2007; Vemprala et al., 2024; Zhang et al., 2023), each of which may recruit different cognitive resources and create different opportunities for misunderstanding (Goodrich & Schultz, 2007). The result is that interaction failures increasingly arise not from what a robot cannot do, but from mismatches in how humans and robots perceive the environment, interpret cues, select actions, and respond, i.e., mismatches that degrade interaction quality. This thesis refers to such mismatches as ‘misalignment’.

As the four studies in this thesis will show, misalignment is not a monolithic phenomenon: it can, for example, take the form of differing visual encoding between human and AI (visual-spatial perceptual misalignment), reference-frame conflicts between operator and robot (spatial cognition-response misalignment), output that superficially appears coherent, but which masks shallow reasoning (verbal response misalignment), or divergent reasoning processes between human and AI that lead to ineffective distribution of cognitive resources (cognitive misalignment). When any of these forms of misalignment is present, the human may need to

invest more effort to sustain the interaction, leading to frustration, errors, and eventual disuse of the technology (Breazeal, 2003; Coronado et al., 2022; Parasuraman & Riley, 1997). The central question of this thesis is therefore: *where do the processes of humans and AI-driven robots align, where do they diverge, and how can interaction be designed so that those differences become complementary?*

1.2 Human and robot information processing

Human information-processing is traditionally modeled through stage-based frameworks, progressing from (1) sensation and perception of the environment, through (2) cognitive decision selection, to (3) response execution (e.g., Wickens, 1992). Artificial agents follow an analogous sequence: namely (1) sensory detection and machine perception, (2) planning and decision selection, and (3) action implementation (e.g., Parasuraman et al., 2000), whether using rule-based symbolic architectures or deep learning paradigms (Laird, 2012; Iosifidis & Tefas, 2022). The underlying assumption is that human and robot partners perceive the same environment through different sensors, select actions through different reasoning processes, and execute responses through different effectors.

To make this three-stage view more precise about where resources overlap between human and robot, this thesis adopts the vocabulary of Wickens' Multiple Resource Theory (MRT) (Wickens, 2002, 2008). The MRT characterizes human information processing not only along the aforementioned processing stages (perception, cognition, and responding) but also across modalities (visual vs. auditory for perception), and processing codes (spatial vs. verbal). Originally, the MRT was developed to predict human performance decrements when two tasks compete for the same resource pool, and to explain efficiency in multitasking (Wickens, 2002). In this thesis, the MRT terminology is used to describe how studies in the thesis are distributed across the information-processing stages and modalities.

Despite decades of relevant work on function allocation (Sheridan & Verplank, 1978; Parasuraman et al., 2000), mutual understanding in communication (Clark & Brennan, 1991), situation awareness (Endsley, 1995), trust calibration (Parasuraman & Riley, 1997), and more recently robot self-explanation (Wallkötter et al., 2021) and robot learning with multimodal feedback (Habibian et al., 2025), the field still lacks an integrated account of how human and robot align or misalign across the full information-processing chain (Xia et al., 2025). This thesis addresses that gap by examining where alignment breaks down and how interaction roles can be distributed for better alignment.

1.3 Research questions and thesis content

To address the central question posed in Section 1.1, this thesis investigates alignment across four studies that span different interaction modalities, robot platforms, and methods. The four studies progress from controlled benchmarking of non-embodied AI on isolated tasks, through laboratory teleoperation and multimodal dialogue with mobile and manipulator robots, to real-world educational deployment of human-AI cognitive collaboration.

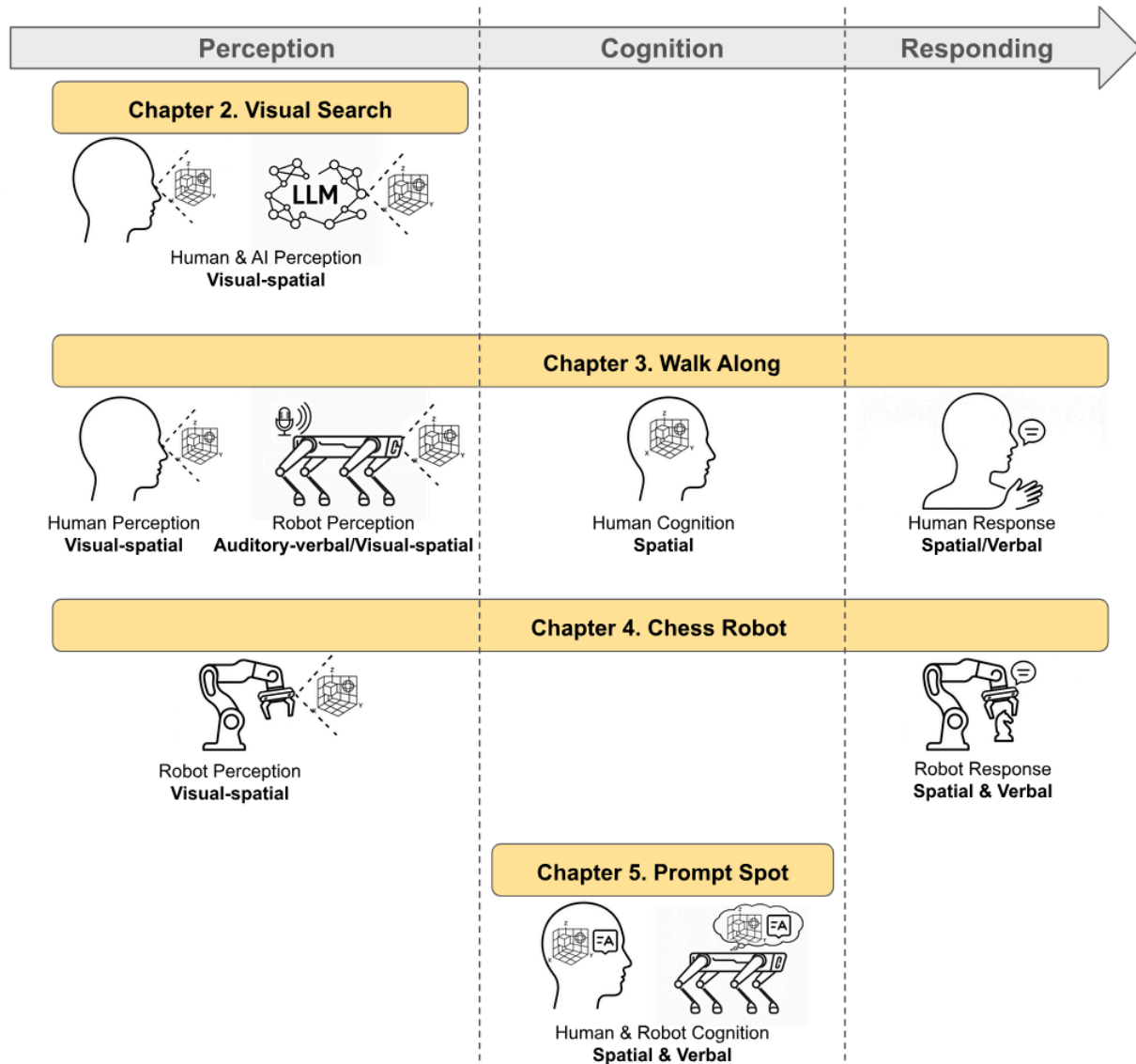


Figure 1.1. Mapping of the four thesis chapters onto the dimensions of Wickens' multiple resource theory (MRT) terminology. The yellow bars indicate which stages of information processing each chapter covers, while the icons highlight the specific components that are explicitly studied or developed in that chapter. The information-processing stages are: (1) perception, (2) cognition, and (3) responding. Furthermore, perception in this thesis can involve the visual or auditory modalities, and processing code can be spatial or verbal.

Research question 1 (Chapter 2): *Do MLLMs exhibit human-like attentional bottlenecks in visual search, and where does their processing diverge from human scanning strategy?*

Human-AI collaboration presupposes at least a partial correspondence between what each agent perceives. However, similar perceptual outcomes do not guarantee similar underlying processes. Chapter 2 isolates the perception stage of visual-spatial processing. Human visual processing operates using two channels: focal and ambient (Wickens, 2002). Like human ambient processing, MLLMs encode the full scene in a single parallel pass. The critical difference is that humans then deploy strategic focal fixations and continue searching serially until a quitting threshold is reached.

The chapter compares human and MLLM performance on identical visual search tasks that vary in difficulty, set size, and dimensionality (2D and 3D). Human and MLLM error patterns turn out to be strongly correlated overall, but diverge in complex conditions: humans trade speed for accuracy and continue searching via focal fixations until a quitting threshold is reached, whereas MLLMs, processing in a single ambient-like pass, tend to default to a binary present-or-absent judgment in difficult conjunction searches. This dissociation between outcome-level alignment and mechanism-level divergence is relevant to the overarching research question, because it defines the most elementary level at which misalignment can arise: two agents that appear to perceive the same scene may in fact fail under different conditions.

Research question 2 (Chapter 3): *How do control modality (voice vs. gestures) and user mobility (standing vs. walking) jointly affect task performance and perceived intuitiveness when directing a mobile robot?*

This chapter moves to the interaction itself: how the human communicates with a robot determines which cognitive resources are recruited, and thereby where misalignment may occur. The study spans all three processing stages: the operator monitors the robot visually (perception, visual-spatial), maintains a spatial model of the robot's pose and selects the next command (cognition, spatial), and issues commands through either speech (response, verbal) or gesture (response, spatial). On the robot side, a custom gesture recognition module was developed to receive these commands through vision (visual-spatial), while speech commands were captured through the HoloLens built-in speech recognition (auditory-verbal). Task performance therefore depends on two factors: whether the robot correctly detects the command (a technical recognition issue) and whether the command-to-movement mapping is intuitive for the operator (a human factors issue). For the human operator, gestures share the spatial processing already used for visual-spatial monitoring, whereas verbal-coded voice commands do not compete for the same resources (Wickens, 2002, 2008).

In a controlled experiment, 218 participants teleoperate the quadruped robot Spot using voice or gesture commands while either standing in a fixed position or walking alongside the robot. Voice

control combined with walking emerges as the most favored and intuitive condition, while gestures issued from a stationary position cause systematic left-right confusion due to spatial misalignment between operator and robot. Chapter 3 contributes to the overarching research question by demonstrating that misalignment is not just a property of the agents themselves but also of the interface: changing the command modality redistributes cognitive load across the operator's resources, with measurable consequences for spatial reasoning and task performance.

Research question 3 (Chapter 4): *Chapter 4 presents a design study with a research component: an open-source, reproducible robotic platform integrating visual perception, AI-generated commentary, and multimodal (verbal and gestural) interaction as a testbed for HRI. The research question it addressed is: how do the robot's physical task execution and verbal commentary jointly shape the robot's perceived usefulness among general and expert users?*

Chapters 2 and 3 each isolate specific information-processing stages, yet realistic collaboration engages perception, reasoning, and communication simultaneously. Investigating alignment across these dimensions requires a platform that is controlled enough for precise measurement yet sufficiently complex to reveal where coordination succeeds and where it fails. Chess satisfies both criteria: it is rule-bounded with objectively verifiable outcomes, and human and AI chess reasoning differ in well-documented ways (McCarthy, 1990), with humans relying on pattern recognition and positional intuition (Chase & Simon, 1973) while AI employs tree search with either hand-crafted (Shannon, 1950) or learned (Silver et al., 2018) evaluation.

Chapter 4 is primarily a design study with a research layer. As shown in Figure 1.1, the platform covers the perception and response stages on the robot side: the robot perceives the board visually and responds through both physical piece manipulation and LLM-generated spoken commentary. An international survey ($N = 597$) and an expert gameplay session reveal that the platform is valued in educational contexts for its capacity to make the robot's reasoning accessible through verbal explanations, while also exposing the limitations of current LLMs in providing the kind of strategic depth that expert players expect. The chapter contributes to the overarching research question by providing the research infrastructure needed to observe alignment and misalignment when multiple information-processing stages are engaged concurrently, and by revealing a concrete instance of verbal response misalignment: LLM-generated commentary that is fluent yet lacks strategic depth, illustrating how coherent output can mask shallow reasoning.

Research question 4 (Chapter 5): *Can a scalable, gamified educational workshop teach engineering students to recognize cognitive limitations in different LLM types and to redistribute reasoning effectively between themselves and the model across tasks of increasing complexity?*

Chapters 2 through 4 characterize where human and AI cognition align and misalign across perception, communication, and multimodal interaction. A recurring implication is that effective collaboration requires the human not only to interact with the AI but to recognize where its reasoning is limited and to compensate accordingly. This chapter addresses that cognitive dimension. As shown in Figure 1.1, the study is located at the cognition stage, engaging both spatial and verbal processing codes: students formulate verbal prompts that encode spatial navigation goals and evaluate the spatial plans that LLMs produce. In a series of workshops ($N = 226$), students prompt a non-reasoning and a reasoning LLM to solve grid-based navigation tasks of increasing complexity, with plans executed in simulation and demonstrated on a Boston Dynamics Spot robot. As tasks grow harder, students transition from providing step-by-step instructions to offering higher-level guidance such as waypoints and strategic heuristics, effectively learning to redistribute cognitive effort between themselves and the model based on each model's planning capacity. The chapter addresses the “complementary” component of the overarching research question: understanding the cognitive profile of an AI partner enables the human to structure the collaboration so that differences in reasoning become productive.

1.4 Summary

As robots powered by LLMs enter shared human spaces, misalignment becomes a risk: failures arising not from insufficient robot capability but from mismatches in how human and robot perceive, reason, and respond. This thesis includes four studies that span the space of processing stages, modalities, and codes. Chapter 2 isolates the perceptual level and reveals that strongly correlated error patterns between humans and MLLMs coexist with fundamentally different processing mechanisms, with error patterns diverging particularly in complex conditions, defining the most elementary form of misalignment. Chapter 3 spans all three information-processing stages and demonstrates that the operator's choice of command modality redistributes cognitive load across resources, identifying the interface as a locus of misalignment distinct from the agents themselves. Chapter 4 provides a multimodal chess robot platform in which alignment and misalignment become observable across perception and response simultaneously, and reveals a verbal response misalignment in which fluent LLM commentary obscures its shallow strategic understanding. Chapter 5 addresses the cognitive stage: engineering students learn, through prompted navigation tasks, where LLM reasoning diverges from their own and how to redistribute effort accordingly, turning cognitive differences into a collaborative asset. Together, these studies progress from isolated perception through embodied interaction to real-world educational deployment. Chapter 6 synthesizes findings across all four studies, revisits the research questions, and outlines directions for future work.

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